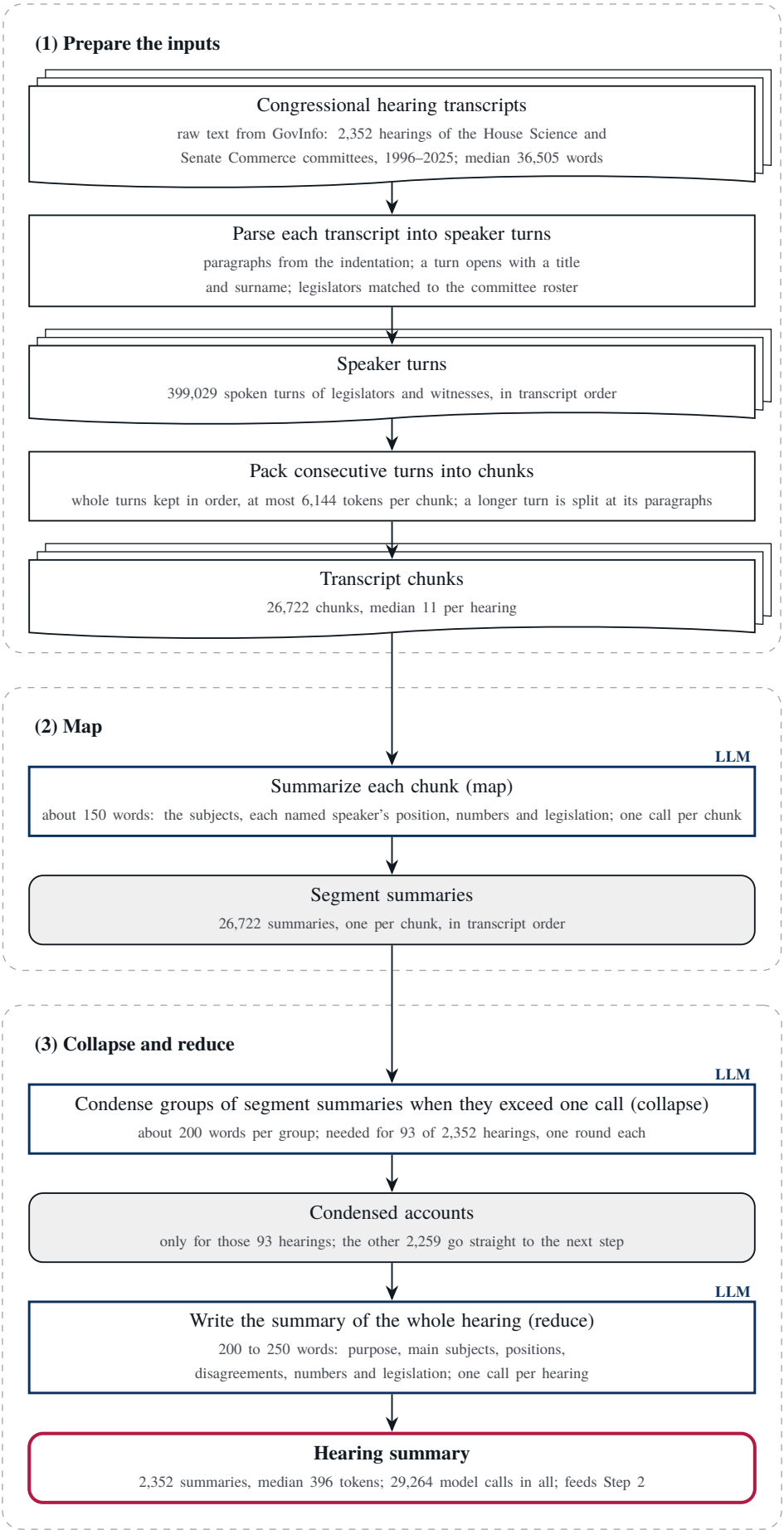

(1) Prepare the inputs
Congressional hearing transcripts
raw text from GovInfo: 2,352 hearings of the House Science and Senate Commerce committees, 1996–2025; median 36,505 words
Parse each transcript into speaker turns
paragraphs from the indentation; a turn opens with a title and surname; legislators matched to the committee roster
Speaker turns
399,029 spoken turns of legislators and witnesses, in transcript order
Pack consecutive turns into chunks
whole turns kept in order, at most 6,144 tokens per chunk; a longer turn is split at its paragraphs
Transcript chunks
26,722 chunks, median 11 per hearing
(2) Map
LLM
Summarize each chunk (map)
about 150 words: the subjects, each named speaker's position, numbers and legislation; one call per chunk
Segment summaries
26,722 summaries, one per chunk, in transcript order
(3) Collapse and reduce
LLM
Condense groups of segment summaries when they exceed one call (collapse)
about 200 words per group; needed for 93 of 2,352 hearings, one round each
Condensed accounts
only for those 93 hearings; the other 2,259 go straight to the next step
LLM
Write the summary of the whole hearing (reduce)
200 to 250 words: purpose, main subjects, positions, disagreements, numbers and legislation; one call per hearing
Hearing summary
2,352 summaries, median 396 tokens; 29,264 model calls in all; feeds Step 2
Raw text documents
Text written by the LLM
Output of the step
LLM step (Qwen3.5-4B)
Computation without a model call